



Alexa routines

An outline of various approaches that can be used to make Alexa work with your smart lights. <https://digit.in/may20-07>



Best smartlights

Whether it is for family rooms, value or security, here are the best smartlights for every purpose. <https://digit.in/may20-08>

This sets the “red color” (PWM1) in the software to full brightness, and all other colours in software to 0. Now, all we need to do is map PWM1 to the different GPIO ports in the template configuration one by one. The bulb will restart after each template change.

STEP 2: Go into template and map PWM1(37) to GPIO4. Make all the other maps go to 0. Click on save. Note what colour the bulb changes to.

STEP 3: Map PWM1(37) to GPIO5. Make all other maps go to 0. Click on save. Note what colour the bulb changes to.

STEP 4: Map PWM1(37) to GPIO12. Make all other maps go to 0. Click on save. Note what colour the bulb changes to.

STEP 5: Map PWM1(37) to GPIO13. Make all other maps go to 0. Click on save. Note what colour the bulb changes to.

STEP 6: Map PWM1(37) to GPIO14. Make all other maps go to 0. Click on save. Note what colour the bulb changes to.

At the end of Step 6, you should have a map similar to this

GPIO4	Green
GPIO5	Red
GPIO12	Yellow
GPIO13	Blue
GPIO14	White

If you don't get all the colour channels you expect from the above GPIO ports, you can try setting random GPIO ports to PWM1 (37), till you find the date for all channels. RGB lights will only have 3 channels, RGBW will have 4 channels, and RGBWY will have 5 channels which need mapping.

Once you have all your data, all you need to do is go into the template configuration and map the right colour channels to the right GPIO pins. You will need to do this according to the following table-

Color	PWM Pin	Code Number
Red	PWM1	37
Green	PWM2	38
Blue	PWM3	39
Yellow	PWM4	40
White	PWM5	41

So, if your GPIO to Color maps were as in the example above, you would get a mapping as follows:

GPIO	PWM	Code Number
GPIO0	None	0
GPIO1	None	0
GPIO2	None	0
GPIO3	None	0
GPIO4	PWM2	38
GPIO5	PWM1	37
GPIO9	None	0
GPIO10	None	0
GPIO12	PWM4	40
GPIO13	PWM3	39
GPIO14	PWM5	41
GPIO15	None	0
GPIO16	None	0
ADC0	None	0

FADE AND SPEED

This is an optional step that allows you set fade and speed options on your bulb. This is optional, but it is highly recommended, as lights with colours that just snap into place can get a little jarring.

Open up the Tasmota console. Enter in the following two commands-

```
Fade ON
Speed 20
```

This will make the bulb fade between settings. You can set speed to any integer between 1 and 20, where 1 is the shortest time between settings, and 20 is the longest.

SLEEP

If you notice your light is flickering when turned on, you can try setting Sleep to a less aggressive setting. Your light will consume slightly more power when turned off, but the trade off in terms of quality of lighting is worth it.

Sleep 0

Once you have HASS properly set up, you can automate your light to set Sleep to 0 when its on for the right behaviour, and set Sleep to 200 for maximum power savings whenever it turns off.

SetOption19

Normally, you would have to manually configure your light and HASS to work together, but Tasmota provides a handy option to play nice with HASS when MQTT autodiscovery is enabled. Head to your Tasmota web UI, and click on console. Type in SetOption19 ON. Now, when you restart your setup, everything will just magically work together! If everything has gone according to keikaku, you will end up with a nice bulb interface in your HASS UI.

BONUS : AUTOMATE SLEEP

If you're experiencing weird flickering, you can automate changing the sleep setting based on the state of the light, to get maximum saving when the light is off, and maximum performance when the light is on. In your HASS interface, head to Configuration -> Automations. Click on the new button on the bottom right corner. On the resulting dialogue box, click “skip”. Name your automation something informative like “PowerOnSleep0”. As your trigger, select “State”. In the entity, select your light. Enter From as “off”, and To as “on”. As your action, select “Call Service”. Select your service to be mqtt.publish. In the payload, enter the following :

```
topic: Tasmota/cmnd/SLEEP
payload: 0
```

Hit save. Add another automation, with the opposite trigger, and payload: 200. Note : The topic will change based on your Tasmota MQTT settings. Tasmota/cmnd/[command name] is the default topic. 